

Quasi-transitivity $\Rightarrow$ Acyclicity

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Quasi-transitivity.** Strict preference is transitive: if $p \succ q$ and $q \succ r$ then $p \succ r$.

0.2 Conclusion

- **Acyclicity.** There is no finite cycle of strict preferences $p_1 \succ p_2 \succ \dots \succ p_n \succ p_1$.

0.3 Proof

If $p_1 \succ p_2 \succ \dots \succ p_n \succ p_1$ then by induction on n , transitivity of $\succ$ gives $p_1 \succ p_n$, and with $p_n \succ p_1$ this contradicts asymmetry of $\succ$.

Record: topics/decision-theory/results/quasi-transitivity-implies-acyclicity.yaml